Name _____

# Unit 7 Summary: Probability Distributions KEY

## Random Variables and Distributions

Flip a coin 2 times. Let $X$ = the number of heads.

| $x$ (heads) | 0 | 1 | 2 |
|---|---|---|---|
| $P(x)$ | $\frac{1}{4}$ | $\frac{2}{4} = \frac{1}{2}$ | $\frac{1}{4}$ |

| **Idea** | **What it means** | **Example: flip a coin 2 times** |
|---|---|---|
| Random variable | A number that comes from a chance process. | $X$ = the number of heads. |
| $X$, $x$, and $P(x)$ | Capital $X$ is the random variable. Small $x$ is one of its values. $P(x)$ means the probability that $X$ equals $x$. | $x$ is 0, 1, or 2. $P(1) = \frac{1}{2}$. |
| Probability distribution | It lists every value of $X$ and its probability. It can be a table or a graph. | The table above. |
| Valid distribution | 1. Every probability is from 0 to 1.<br>2. The probabilities add to 1: $\Sigma P(x) = 1$ ($\Sigma$ means add up). | $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. Valid. |
| Using a distribution | Add the probabilities of the values you want. | $P(X \geq 1) = P(1) + P(2)$<br>$= \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ |

**Symbols:** $\geq$ means "greater than or equal to." $\leq$ means "less than or equal to."

## Theoretical or Empirical?

| **Kind** | **Where it comes from** | **Example** |
|---|---|---|
| Theoretical | Reasoning about the chance process (equally likely outcomes, or the multiplication rule). | The coin flips above. |
| Empirical | Data. Divide each count by the total. | 50 families: 0 cars: 5, 1 car: 20, 2 cars: 18, 3 cars: 7. So<br>$P(0) = \frac{5}{50} = 0.10$, $P(1) = 0.40$,<br>$P(2) = 0.36$, $P(3) = 0.14$. |

## Expected Value

The number of goals a soccer team scores in a game:

| $x$ (goals) | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.1 | 0.3 | 0.3 | 0.2 | 0.1 |
| $x \cdot P(x)$ | 0 | 0.3 | 0.6 | 0.6 | 0.4 |

| Tool | How it works | Example: goals in a game |
|---|---|---|
| Expected value | $E(X) = \Sigma\, x \cdot P(x)$. $E(X)$ means the expected value of $X$. Multiply each value by its probability. Then add them all up. | $E(X) = 0 + 0.3 + 0.6 + 0.6 + 0.4$ = 1.9 goals |
| Mean of $X$ | The expected value is also called the **mean** of $X$, written $\mu_X$ ($\mu$ is "mu"). | $\mu_X$ = 1.9 goals |
| What it means | A **long-run average**. It tells you what happens over many tries. It does not tell you what happens on one try. | Over many games, the team scores about 1.9 goals per game on average. No single game has 1.9 goals. |

## Fair or Not?

| | What it means | Example |
|---|---|---|
| Fair method of choosing | Every person has the same probability of being chosen. | Roll a number cube. 1 or 2: Ana, 3 or 4: Ben, 5 or 6: Cai. Each has $\frac{2}{6} = \frac{1}{3}$. **Fair.** |
| Not a fair method | Not every person has the same probability. | Flip 2 coins. HH: Ana, TT: Ben, one of each: Cai. Ana $\frac{1}{4}$, Ben $\frac{1}{4}$, Cai $\frac{2}{4} = \frac{1}{2}$. **Not fair.** |
| Fair game | The expected value of the money you win or lose is **0**. On average, no one wins or loses money. | Roll a number cube. You win \$3 if you roll a 6. You pay \$1 for any other number. $E(X) = 3 \cdot \frac{1}{6} + (-1) \cdot \frac{5}{6} = -\frac{2}{6} \approx -\$0.33$ You lose about 33 cents a game. **Not fair.** |

## Expected Payoff

A lottery ticket costs \$2. 1 in 1,000 tickets wins \$500. 1 in 50 tickets wins \$10. The rest win nothing.

| Outcome | Net gain $x$ | $P(x)$ | $x \cdot P(x)$ |
|---|---|---|---|
| Win \$500 | $500 - 2 = 498$ | 0.001 | 0.498 |
| Win \$10 | $10 - 2 = 8$ | 0.02 | 0.16 |
| Win nothing | $-2$ | 0.979 | $-1.958$ |
| | $E(X) = \Sigma\, x \cdot P(x)$ | | $-1.30$ |

| Tool | How it works | Example: the lottery ticket |
|---|---|---|
| Net gain | Net gain = what you win − what you paid. It is negative when you lose money. | Win \$500: $500 - 2 = 498$. Win nothing: $-2$. |
| Expected payoff | Let $X$ = the net gain. Then $E(X) = \Sigma\, x \cdot P(x)$. | $0.498 + 0.16 + (-1.958) = -1.30$ |
| What it means | The amount you win or lose on average. | On average, a buyer **loses \$1.30** per ticket. The lottery keeps that money. |

**Negative expected payoff:** you lose money on average. **Expected payoff of 0:** the game is fair.

## Protection Plans

| Step | What to do | Example: a phone |
|---|---|---|
| Expected cost with no plan | Multiply the chance it breaks by the cost to replace it. | A phone costs \$300 to replace. The chance it breaks this year is 0.10. $0.10 \cdot 300 = \$30$ |
| Compare | Compare it to the cost of the plan. | The plan costs \$40 for the year. On average, the plan costs \$10 more. That is how the company makes money. |
| Think about risk | Expected value is one part of a decision. Risk is another part: could you pay the big bill if it came? | The plan protects you from a sudden \$300 bill. If you could not pay it, the plan might be worth it to you. |